

VPN Provisioning Portal

Workflows, system-to-system data flows and decision trees — Catalyst 8000v IPsec VTI + eBGP lab · Nautobot · Cisco Network-as-Code (Terraform) · Robot Framework

Generated from the codebase in github.com/dcantor/cat8000v-ipsec. Every diagram describes what the code actually does (file and function names are given), not an idealised design.

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1. Systems and where the data lives

System	Runs where	Holds (system of record for)	Talks to
VPN Provisioning Portal FastAPI + single-page UI, <code>webapp/</code>	Lab host, <code>systemd --user lab-webapp</code> , port 8090	Run records (<code>webapp/runs/<id>.json</code> + the intent each run deployed); nothing else — it orchestrates	Browser (HTTP/JSON), every tool below via <code>./lab.sh</code> subprocesses, Nautobot (REST/GraphQL), headends (SSH, for the inventory)
lab-intent.json <code>nautobot/intent.py</code>	Lab root (git)	The VPN service intent: site/VPN metadata, regions, devices (hostname, mgmt IP, role, AS, router-id, LAN, region, site, site code, contact, per-spoke PSK), links, tunnels, crypto profile, capacity	Read by seed, renderer (PSKs), tests, portal; written by the portal and the spoke/hub/remove steps
lab.conf + libvirt <code>lab.sh</code>	Lab host, <code>qemu:///system</code>	VM facts: node names, node index (MACs, UDP ports), console ports, management IPs, physical wiring (<code>LINKS</code>), tunnel numbers, hub/spoke port counts; domain XML, overlay disks, day-0 ISOs under <code>nodes/</code>	libvirt (virsh), routers' serial consoles (raw TCP, <code>tools/console.py</code>)
Nautobot 3.2 (+ bgp-models, golden-config, device-onboarding, Gitea) NMS VM 10.0.0.10	Docker Compose on the NMS jumphost	Source of truth: locations (lab site → Region → Branch), devices, interfaces, cables, prefixes/IPs with roles, core VPN app objects (VPN, tunnels, endpoints, profile, Phase 1/2 policies), BGP AS/routing instances/peerings, config contexts, custom fields, saved GraphQL query, Golden Config settings/compliance; Gitea holds backups, intended configs, Jinja templates	Routers (SSH for onboarding/backups), seed/renderer/portal (REST + GraphQL), Gitea (git)

Network-as-Code nac/ : module netascode/nac- iosxe, provider CiscoDevNet/iosxe 0.15	Lab host, Terraform CLI	global.nac.yaml (hand-written baseline), generated devices.nac.yaml and device_groups.nac.yaml (PSK variables), Terraform state (terraform.tfstate)	Routers over RESTCONF (HTTPS 443, YANG native models); a cisco- ia:save-config RPC after apply
Catalyst 8000v routers	libvirt VMs on the lab host	Running/startup configuration; live IKEv2/IPsec/BGP state; platform CPU / DRAM (capacity dashboard)	Each other (UDP-tunnelled p2p links, IPsec VTIs, eBGP), NMS via OOB network 10.2.0.0/24 (Mgmt-vrf)
VyOS firewalls fw-east / fw-central / fw-west 1 vCPU / 1 GiB, overlays on images/vyos- base.qcow2	libvirt VMs on the lab host, one in front of each headend	Running configuration rendered from Nautobot (nautobot/render_vyos.py): eth1 to the headend, eth2.. to the spokes, forward filter (IKE / ESP / ICMP allowed, default drop + log), LLDP; static bandwidth_mbps in the intent → custom field firewall_bandwidth_mbps	Headend and spokes (UDP- tunnelled links, all WAN /30s terminate on the firewall), NMS via OOB, renderer/tests over SSH (netmiko vyos), console day-0 (tools/vyos_console.py)
Robot Framework tests/	Lab host venv	Results per run under results/<date>_<time>/ (report, log, pre/post config backups, diff)	Routers (SSH via netmiko, RESTCONF), Nautobot (REST/GraphQL), Terraform (plan exit code), renderer (- - check), Gitea (raw backups)

What is deliberately not in Nautobot: pre-shared keys (one per spoke; they live only in lab-intent.json and are rendered as NAC variables into nac/data/device_groups.nac.yaml) and device credentials (Nautobot secrets group lab-devices resolves them from environment variables on the NMS). **What is deliberately not in the intent:** anything discovered from the routers (serials, model, platform) — that comes from Nautobot onboarding.



Edge	Protocol	Data passed
Browser → Portal	HTTP JSON	Intent document; SpokeSpec / HubSpec / removal name; run options (golden, test, delete_disk); polling offsets
Portal → lab.conf / nodes/	file edits (regex on bash arrays)	New [name]= entries in ROLE/MGMT_IP/BGP_AS/LAN/CONSOLE_PORT/NODE_IDX, LINKS "hub:port spoke:port prefix", TUNNELS "id hub spoke prefix", ROUTERS/ALL_NODES; day-0 iosxe_config.txt from nodes/_template/ (hostname, mgmt IP, gateway substituted)
lab.sh → libvirt	virsh (define/start/undefine)	Domain XML: deterministic UUID (sha1 of the management IP → stable C8000v serial), Gi1 on the OOB bridge, GiN as UDP-socket NICs (pairs anchored on the headend: local 22000+idx·100+port , far 32000+idx·100+port), serial console TCP port, CVAC ISO
lab.sh → router console	raw TCP (tools/console.py)	Day-0 config (AAA, Mgmt-vrf/Gi1, SSH, NETCONF/RESTCONF, license boot level), RSA key generation, show version (license check) and a reload when needed
Onboarding → routers	SSH (netmiko/NAPALM inside Nautobot)	Hostname, model, serial, platform, interfaces; then onboard.py refreshes serials itself over SSH (the job does not update existing devices)
seed.py → Nautobot	REST (pynautobot) + GraphQL reads + two ORM writes via nbshell	Location types/locations, custom fields, roles, devices (matched by primary IP), interfaces (+MAC, enabled), IPs/prefixes (+roles, tags), cables, VPN policies/profile/VPN/tunnels/endpoints, AS/routing instances/peerings, config context, saved GraphQL query
Nautobot → renderer	GraphQL (saved query)	Per device: name, role, primary IP, config context, interfaces (type, enabled, IPs with parent prefix role/tags, connected interface, VPN endpoint → profile/policies, tunnel → far endpoint's source address), BGP routing instances with endpoints and peers
Renderer → NAC data	YAML files	devices.nac.yaml : per device system, crypto (proposal/policy/keyring peers/profile/transform set/IPsec profile), ethernet, loopbacks, native tunnels, BGP; CLI templates for ip tcp adjust-mss . device_groups.nac.yaml : group IPSEC_VPN with psk_<spoke> variables
Terraform → routers	RESTCONF (HTTPS, YANG Cisco-IOS-XE-native)	One resource per config object; PATCH on create/update, DELETE on destroy; -parallelism=1 to avoid datastore lock races; cisco-ia:save-config RPC per router after apply
Golden Config → routers / Gitea	SSH (nornir-netmiko), git	Backup: running-config → config-backups/<device>.cfg ; Intended: Jinja2 c8000v-ipsec.j2 rendered from GraphQL (golden-config-lab query) → intended-configs/ ; Compliance: per-feature diff rows (17 features × N devices)
Robot → everything	SSH / RESTCONF / REST / GraphQL / subprocess	Assertions on live state vs the intent (lab_vars.py), on Nautobot objects, on Terraform drift (exit code), on renderer round-trip, on Golden Config compliance
Portal inventory → headends	SSH (netmiko)	show crypto ikev2 sa, show ip interface brief, show interfaces TunnelN, show bgp ipv4 unicast summary, show crypto ipsec sa, show platform resources → per-tunnel health, SA age, BGP state/prefixes, ESP counters, rates; control-plane CPU vs the platform's warning threshold, QFP CPU, DRAM
render_vyos.py → firewalls	SSH (netmiko vyos)	Per firewall from Nautobot (devices with role vpn-firewall, interfaces + cables + addresses, config context firewall policy): delete + set for eth1..eth8 (address, description, unwired = disabled), the forward filter rules, LLDP — in one commit; --check compares the running set lines, --dry-run prints

Headend capacity (portal)	computed	Per headend: tunnels vs <code>vpn_tunnel_capacity</code> ; bandwidth used (tunnels × <code>bandwidth_per_tunnel_mbps</code> from the VPN's extra attributes) vs the firewall's <code>firewall_bandwidth_mbps</code> ; live control-plane CPU vs threshold. Aggregate = the tightest; effective free slots = min of the modelled constraints (0 when CPU ≥ threshold); same math in <code>intent.headend_capacity()</code> for the wizard and validation
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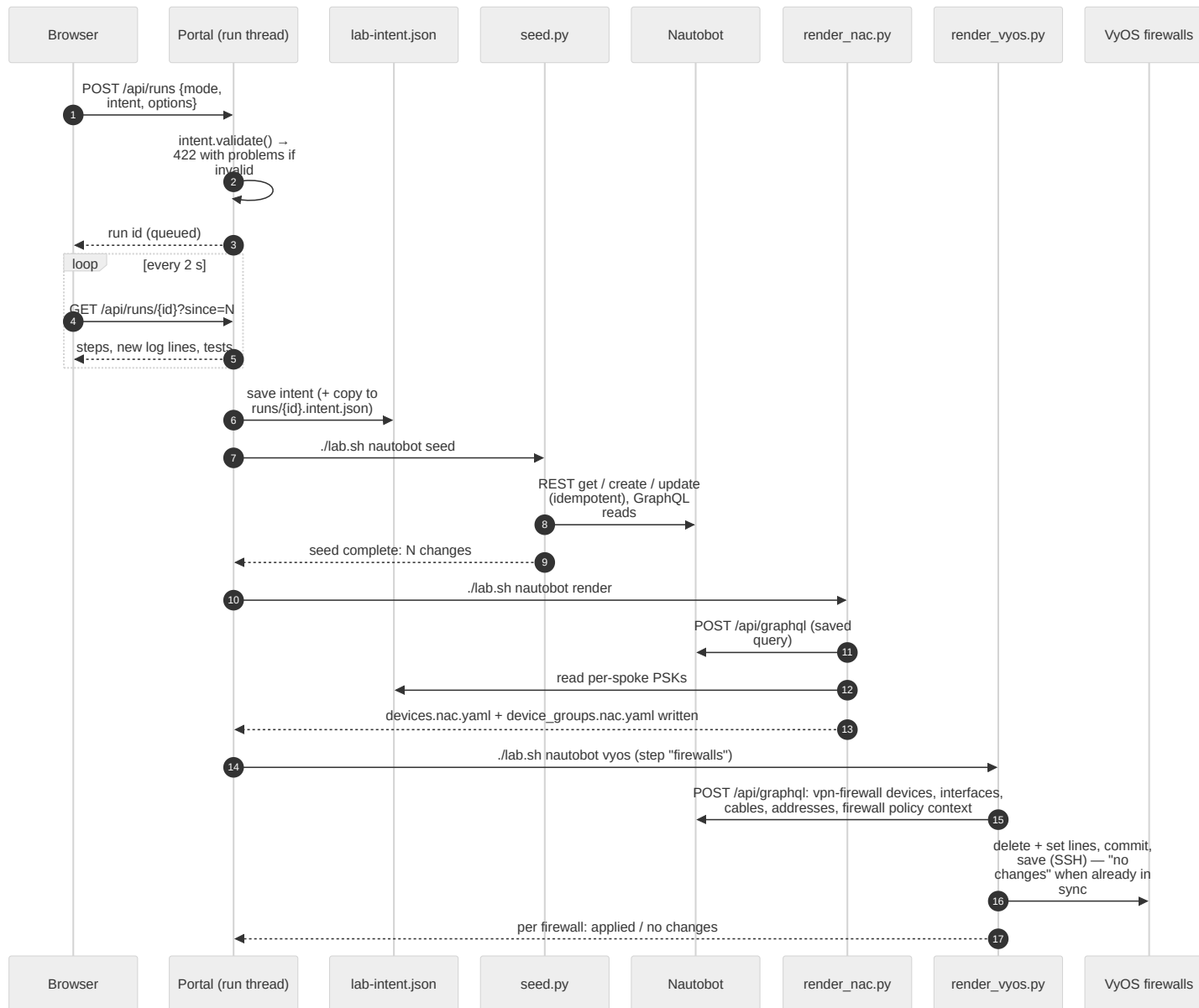
3. The intent document

`lab-intent.json` is the single input to every workflow. It is created once from `lab.conf` (`./lab.sh intent init`) and afterwards only written by the portal (deploy) or by the spoke/hub/remove steps. `intent.load()` upgrades older documents in place (tunnels gain a `hub`, spokes gain a `psk`, devices gain `region/site`).

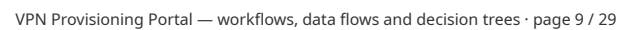
Key	Content	Consumed by
<code>site</code>	Lab-level location name (read-only), description, <code>site_code</code> , <code>contact</code>	seed (location + custom fields), tests (<code>NAUTOBOT_LOCATION</code>), inventory scope, Golden Config dynamic group
<code>vpn</code>	VPN name, description, <code>change_ticket</code> , <code>owner</code>	seed (VPN object + <code>extra_attributes</code>), run records, inventory/CSV
<code>profile</code>	<code>name</code> , <code>ike</code> {encryption, integrity, <code>dh_group</code> , lifetime}, <code>ipsec</code> {encryption, integrity, lifetime}, <code>dpd</code> {enabled, interval, retries}, <code>ios</code> object names	seed (Phase 1/2 policies, profile, <code>extra_options.ios</code>) → renderer/Golden Config read them back from Nautobot, never from the intent
<code>regions</code>	Ordered list; distance between regions = index difference	seed (Region locations), wizard (nearest two headends), inventory grouping
<code>devices[]</code>	<code>name</code> , <code>mgmt_ip</code> , role hub spoke, <code>asn</code> , <code>router_id</code> , <code>lan</code> , <code>comments</code> , <code>region</code> , <code>site</code> , <code>site_code</code> , <code>contact</code> , <code>psk</code> (spokes)	seed (devices matched by mgmt IP → renamed if needed; loopbacks; AS; branch location), renderer (PSK variables), tests, portal tables
<code>links[]</code>	<code>a</code> (headend), <code>a_port</code> , <code>b</code> (spoke), <code>b_port</code> , <code>prefix /30</code> — must equal the physical wiring in <code>lab.conf</code>	validation (wiring check), seed (cables, WAN prefixes/IPs, interface enabled/description)
<code>tunnels[]</code>	<code>id</code> (TunnelN both ends), <code>hub</code> , <code>spoke</code> , <code>prefix /30</code> — one per (hub, spoke)	seed (TunnelN interfaces, VPN tunnel + endpoints, BGP peering), tests
<code>oob</code> , <code>domain_name</code>	Mgmt-vrf, gateway, ACL name, OOB prefix; domain	seed (config context) → renderer/Golden Config via <code>config_context</code>
<code>capacity</code>	<code>tunnels_per_headend</code> (50), <code>bandwidth_per_tunnel_mbps</code> (8)	validation (per hub: tunnel count and firewall bandwidth), seed (custom field on hubs; VPN <code>extra_attributes</code>), inventory bars (tunnels, firewall bandwidth, aggregate), wizard
<code>devices[].bandwidth_mbps</code> (firewalls)	static throughput of the firewall in front of a headend	seed (custom field <code>firewall_bandwidth_mbps</code>), <code>intent.headend_capacity()</code> , inventory, wizard

4. Workflow: Deploy / Dry run

Triggered by **Deploy configuration** (mode `deploy`) or **Dry run** (mode `plan`) on the Provision page, or `POST /api/runs`. Steps: `validate` → `save` → `nautobot` → `render` → `plan` → `apply` → `golden` → `test` (plan stops after `plan`).

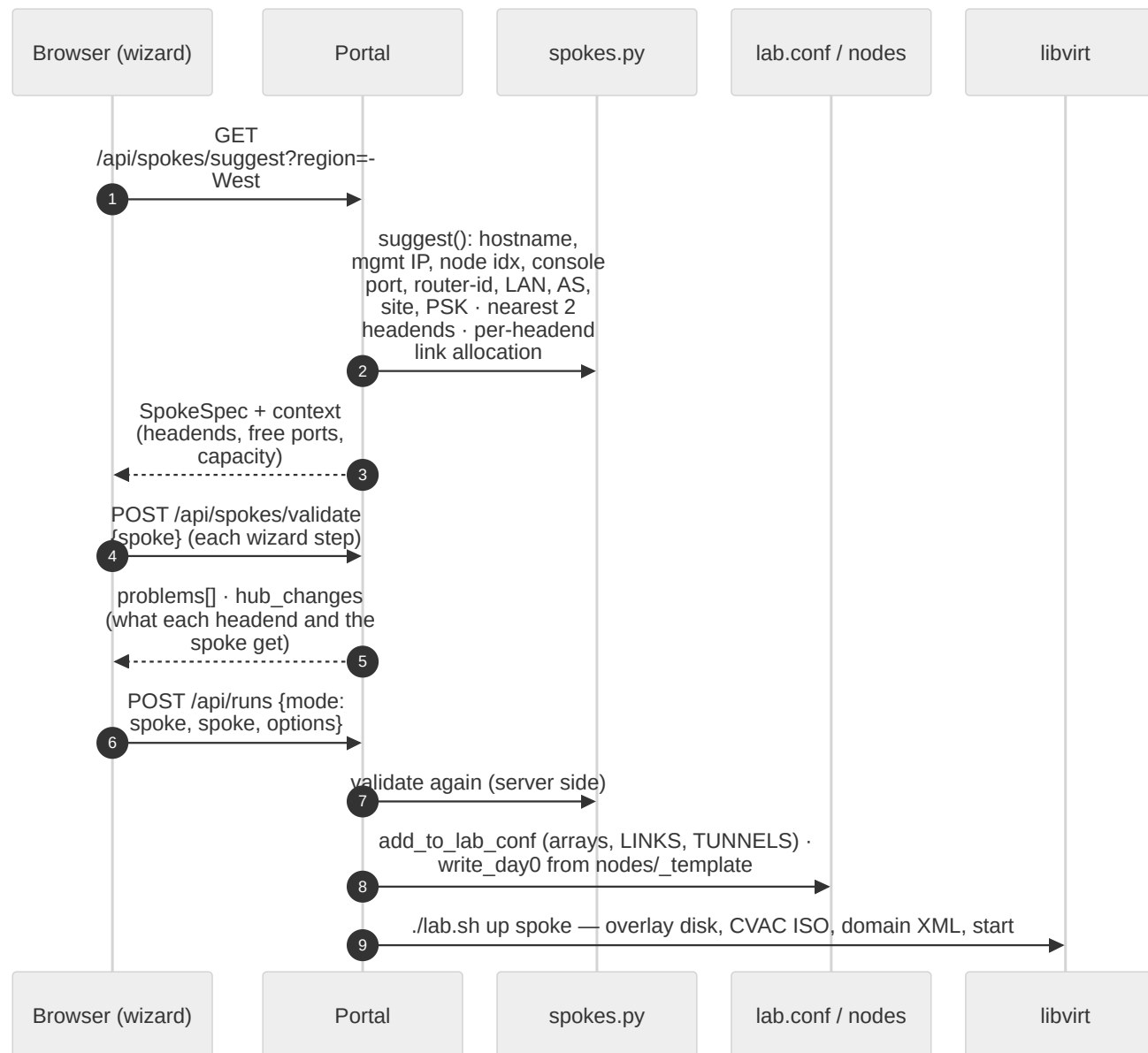


Phase 2 — plan, apply, verify

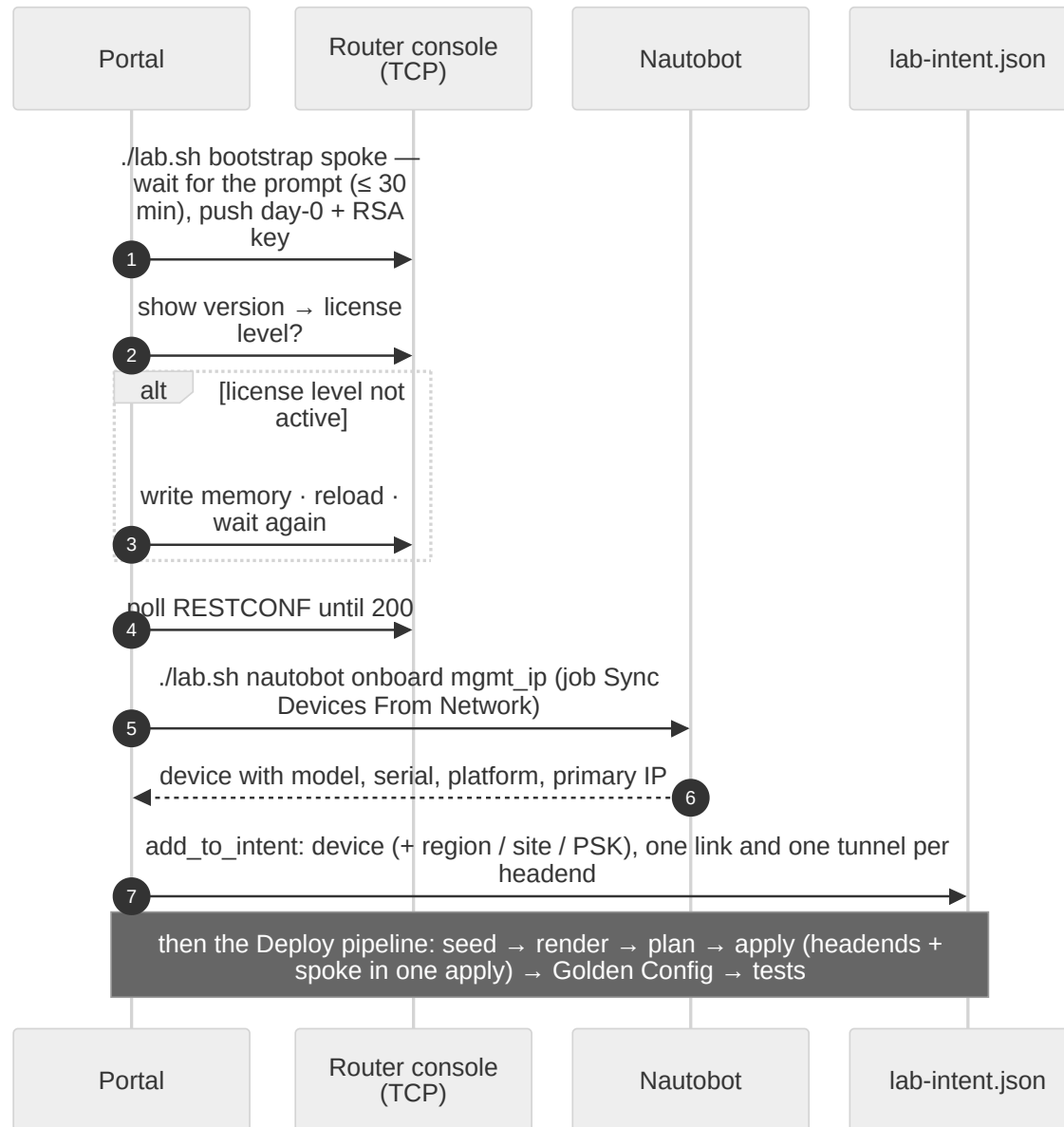


5. Workflow: Add a spoke

The wizard (3 steps) ends in `POST /api/runs {mode: "spoke", spoke: SpokeSpec}`. Steps: `spoke_validate` → `spoke_labconf` → `spoke_vm` → `spoke_bootstrap` → `spoke_onboard` → `spoke_intent` → `nautobot` → `render` → `plan` → `apply` → `golden` → `test`.



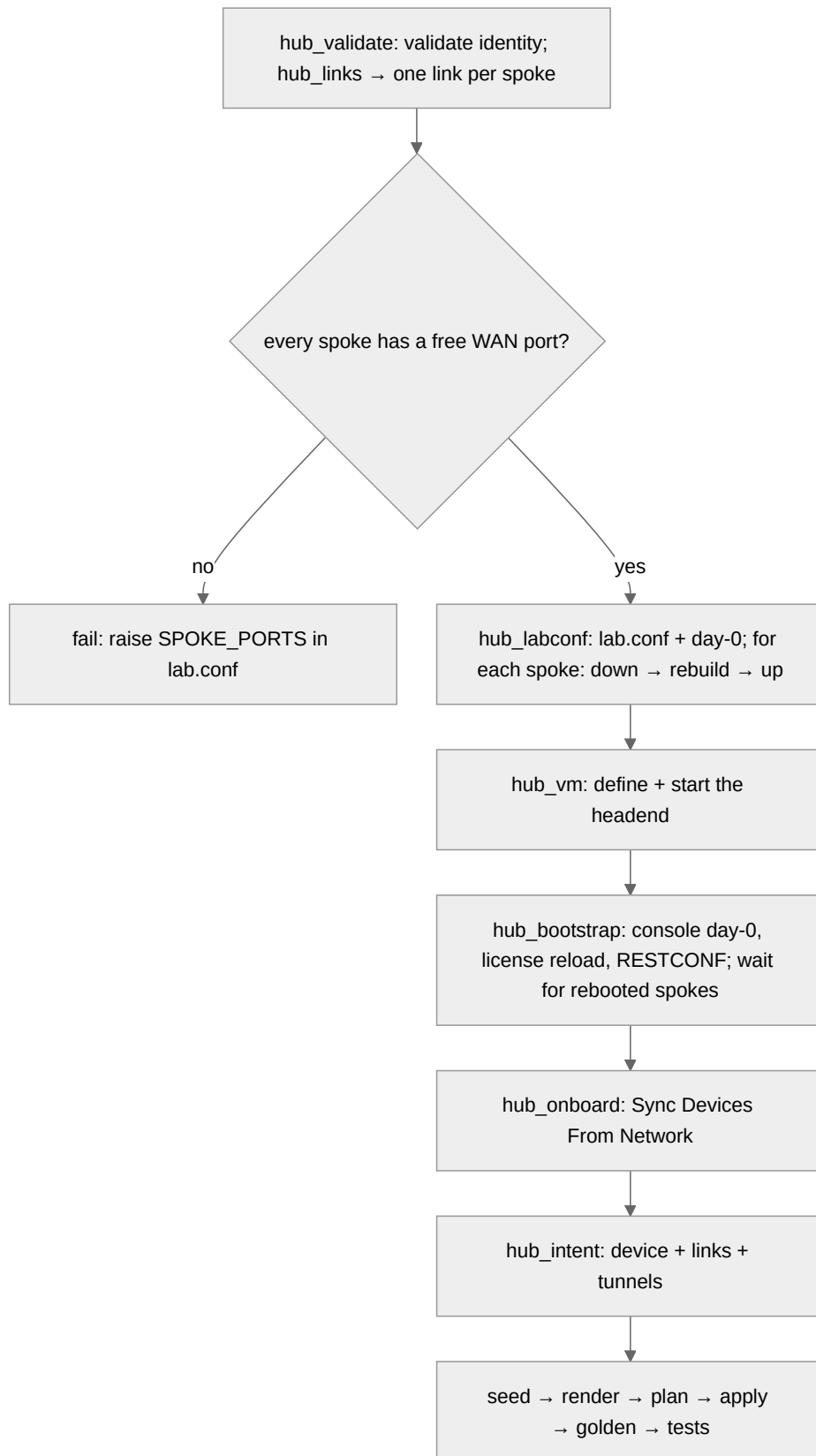
Phase 2 — bring the router up and into the model



Every step is idempotent so a failed run can be **resumed**: `spoke_labconf` skips if the name is already in `lab.conf`, `lab.sh up` reports "already running", `spoke_intent` skips if the device is already in the intent.

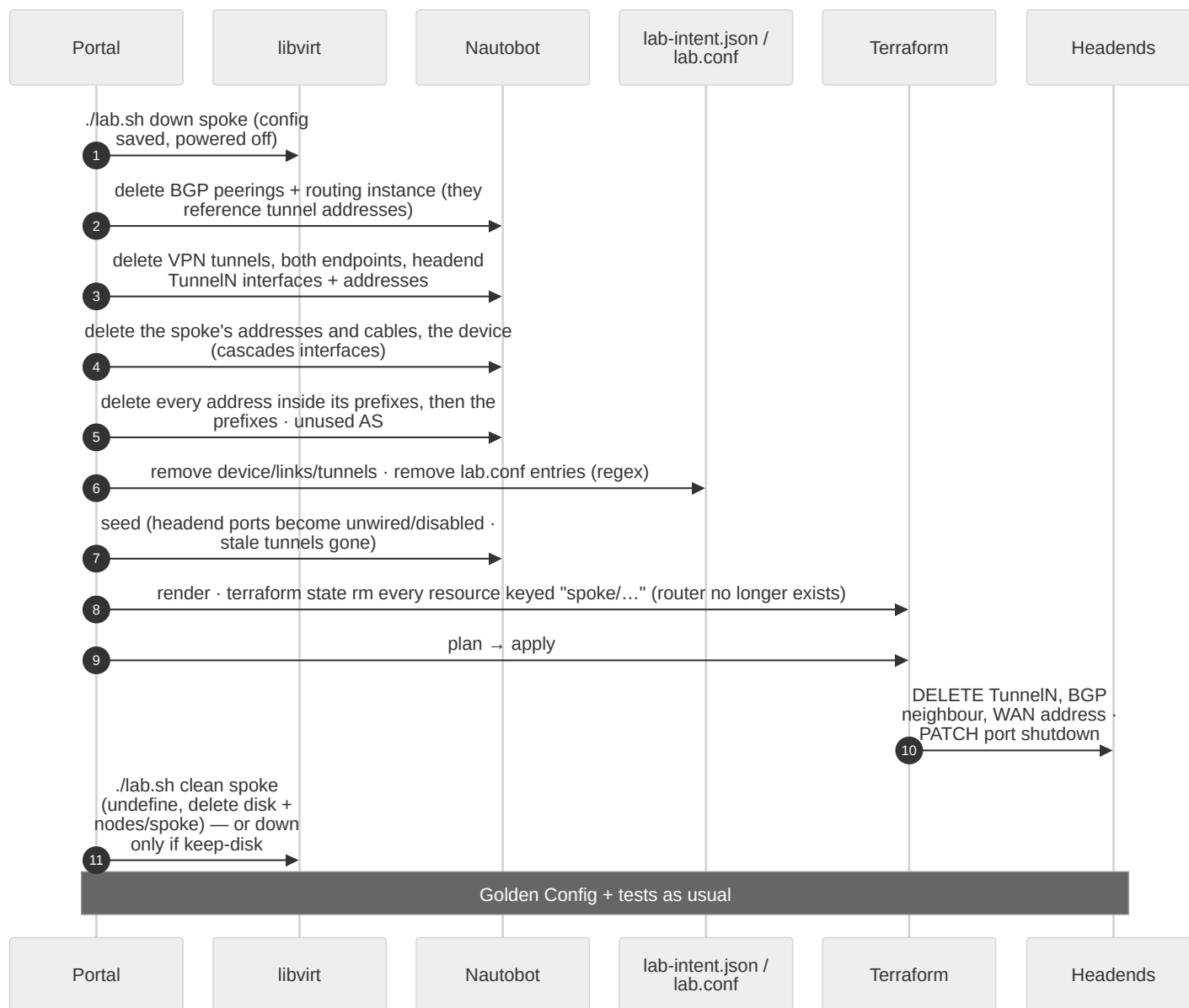
6. Workflow: Add a headend

`POST /api/runs {mode: "hub", hub: HubSpec}` (from `GET /api/hubs/suggest`). Same shape as a spoke, with two differences: the new headend gets a link + tunnel to **every existing spoke** (`spokes.hub_links()`, each spoke's next free WAN port), and because a link is anchored on the headend's UDP ports, each existing spoke's libvirt definition changes — the run re-defines and reboots each spoke (`down / rebuild / up`) and waits for them after the headend's bootstrap.



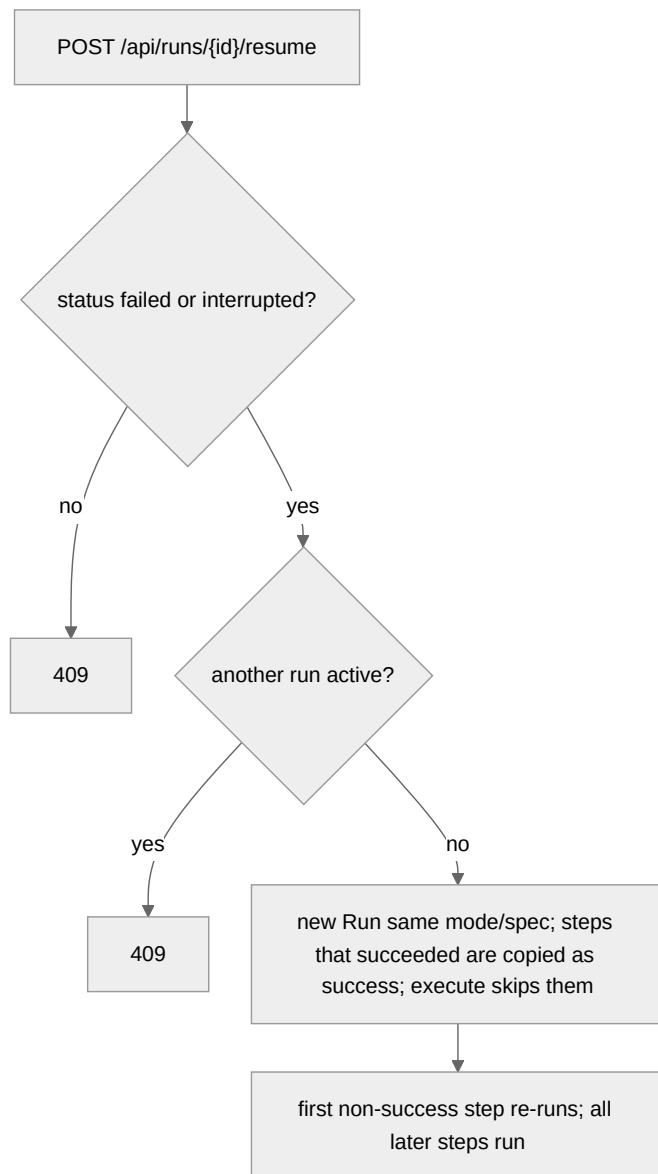
7. Workflow: Remove a spoke

Remove... on a spoke row → GET /api/spokes/{name}/removal (plan dialog) → POST /api/runs {mode: "remove", spoke: {name}}. Steps: rm_validate → rm_down → rm_nautobot → rm_intent → nautobot → render → rm_state → plan → apply → rm_vm → golden → test.



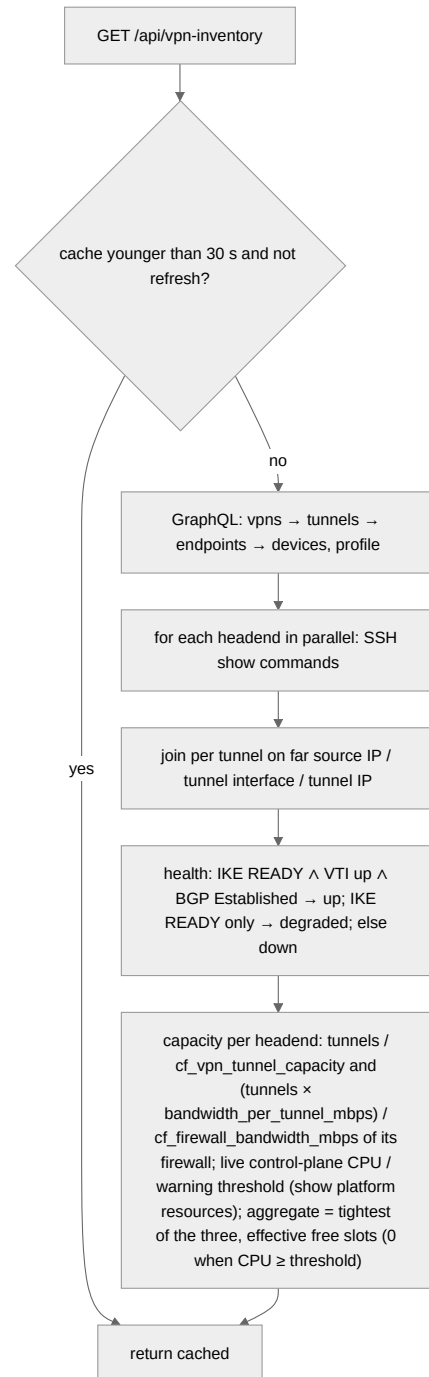
8. Workflow: Resume, tests, inventory

Resume



On start-up the portal marks runs found "running" as **interrupted** (running step → failed, pending → skipped) so they are resumable. `webapp/restart.sh` refuses to restart while a run is active.

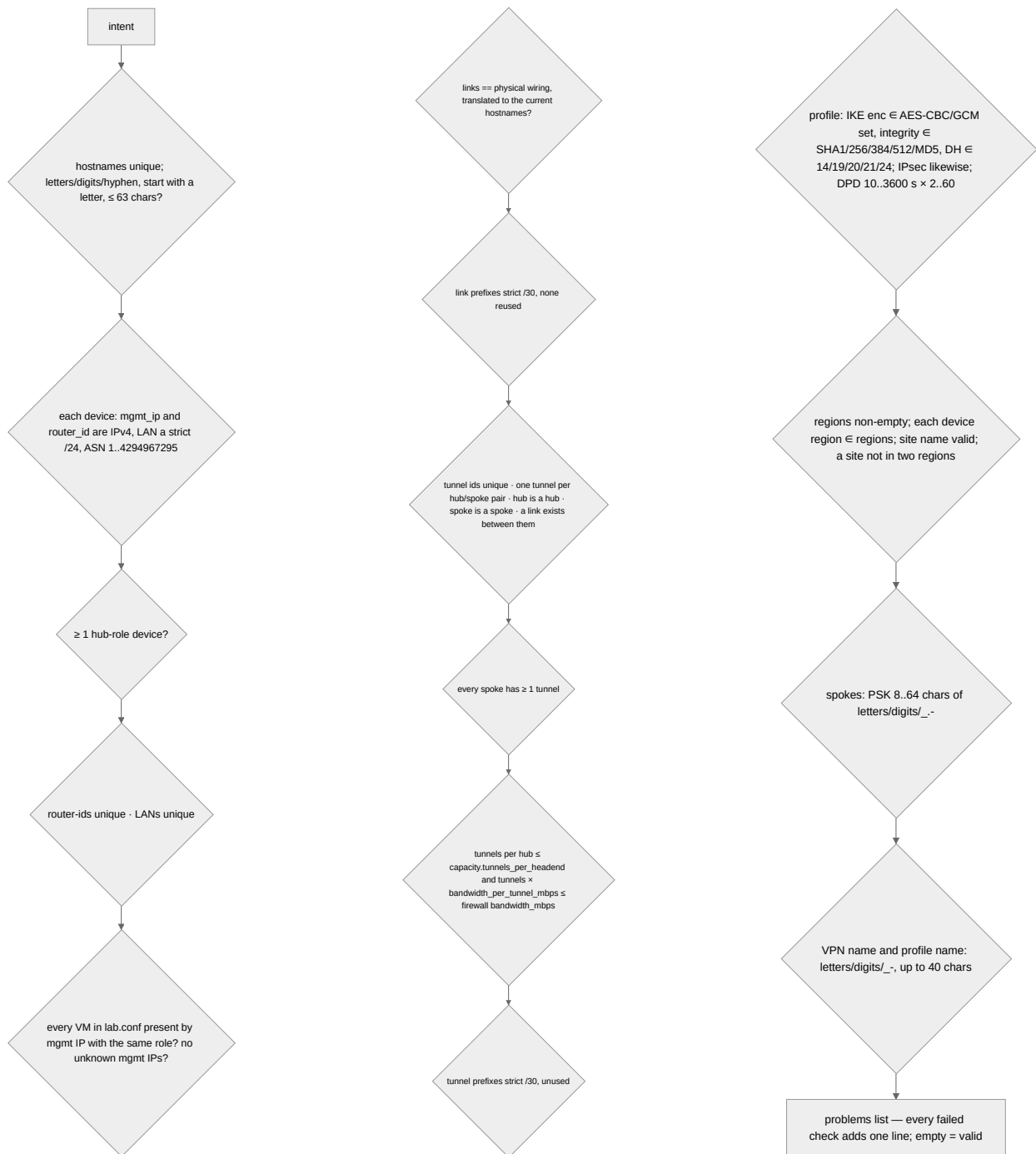
Inventory refresh



Tests (mode test , or the last step of every run)

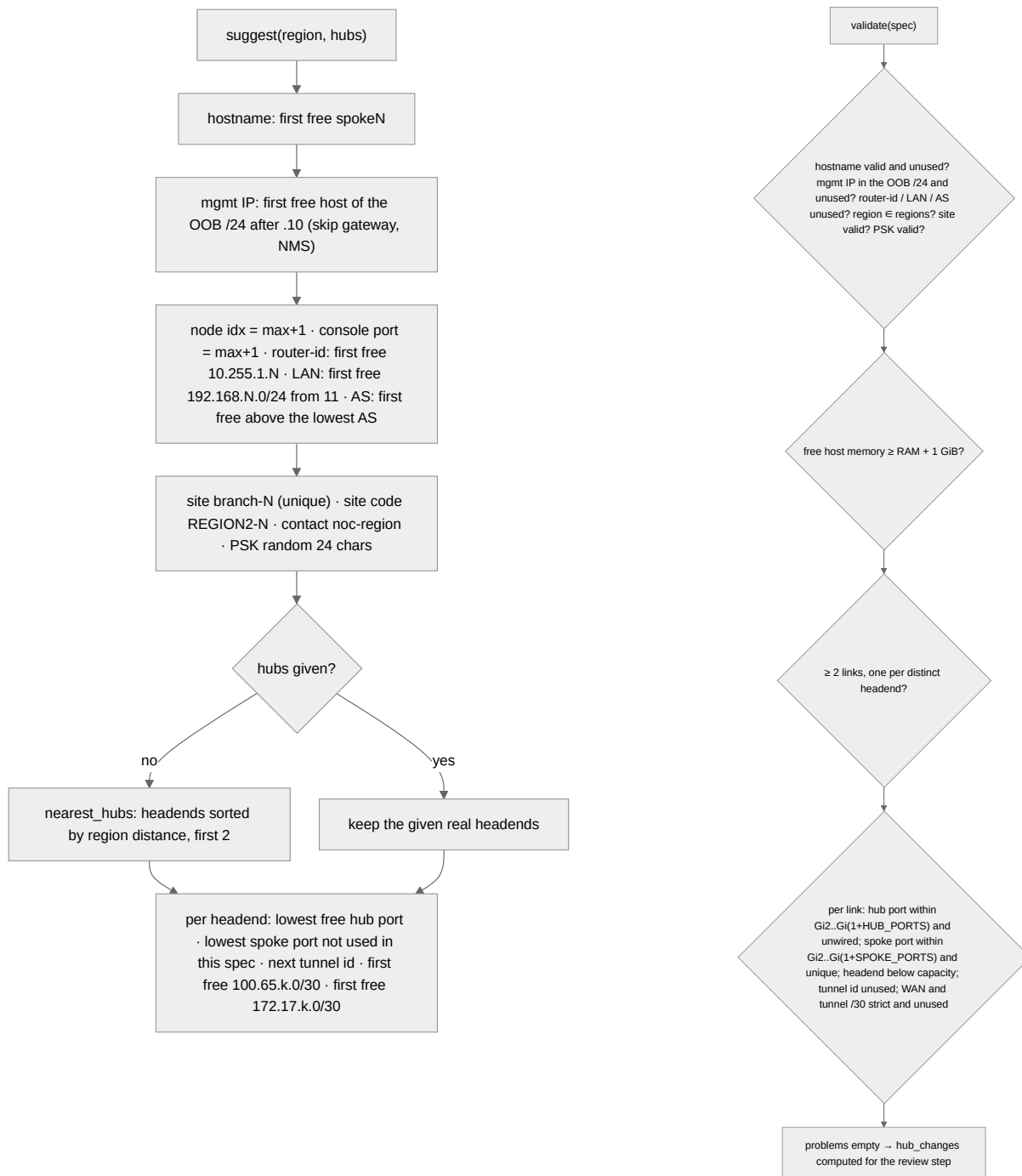
`tests/run.sh` creates `results/<date>_<time>/`, captures every router's running/startup config (`configs/pre-run`), runs the six suites, captures again (`configs/post-run`) and diffs; `results/latest` points at the newest run. The portal parses `output.xml` into the per-suite report shown in the UI and serves `report.html / log.html / backups` under `/results/`.

9. Decision trees: intent validation (`intent.validate`)

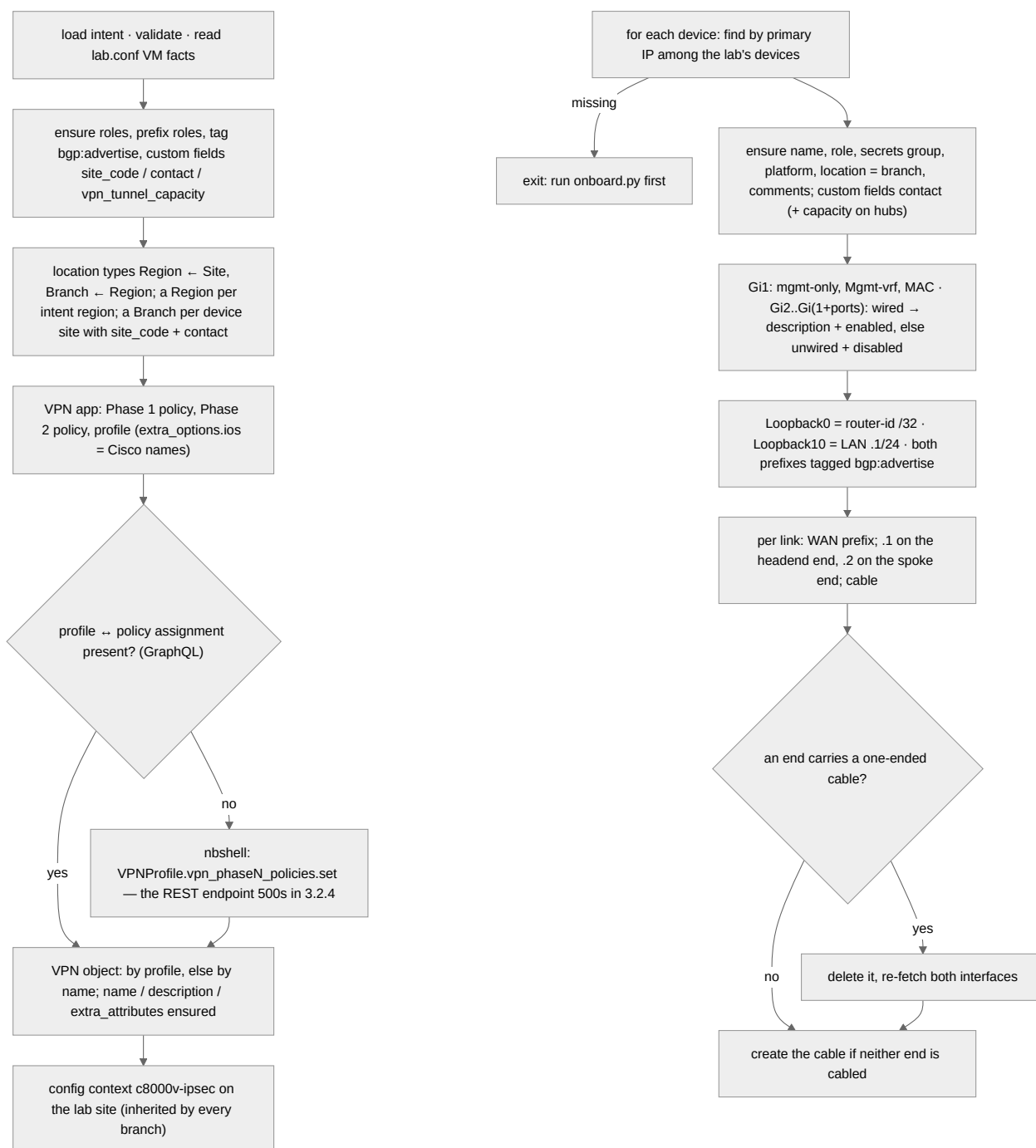


All checks run; the caller gets the full list. The pipeline refuses to start (HTTP 422) or fails its first step on any problem.

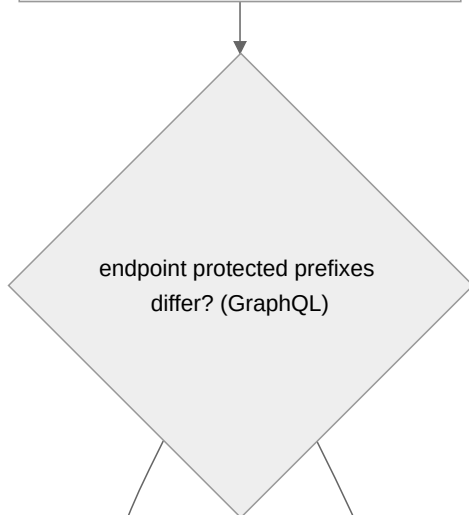
10. Decision trees: spoke allocation (`spokes.suggest` / `validate`)



11. Decision trees: the Nautobot seed (`nautobot/seed.py`)



per tunnel, both ends: delete other TunnelX with the same far-end description; TunnelN (type tunnel) + /30 IP; endpoint (role, source iface + IP, tunnel iface, protected prefixes)



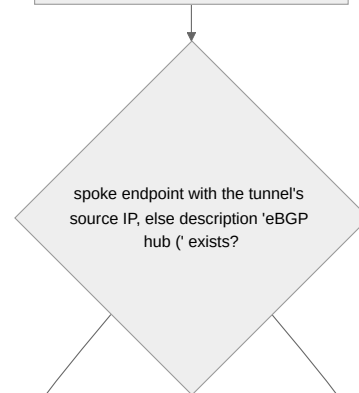
yes
direct PATCH (M2M absent from REST reads)

no

VPN tunnel: by vpn + tunnel_id, else by name; A = headend, Z = spoke; encapsulation IPsec-Tunnel

delete this VPN's tunnels that are not in the intent

BGP: AS per device; routing instance (router-id = Loopback0); ipv4_unicast AF



yes

re-point source IPs / AS / descriptions

no

create peering + both endpoints + endpoint AFs

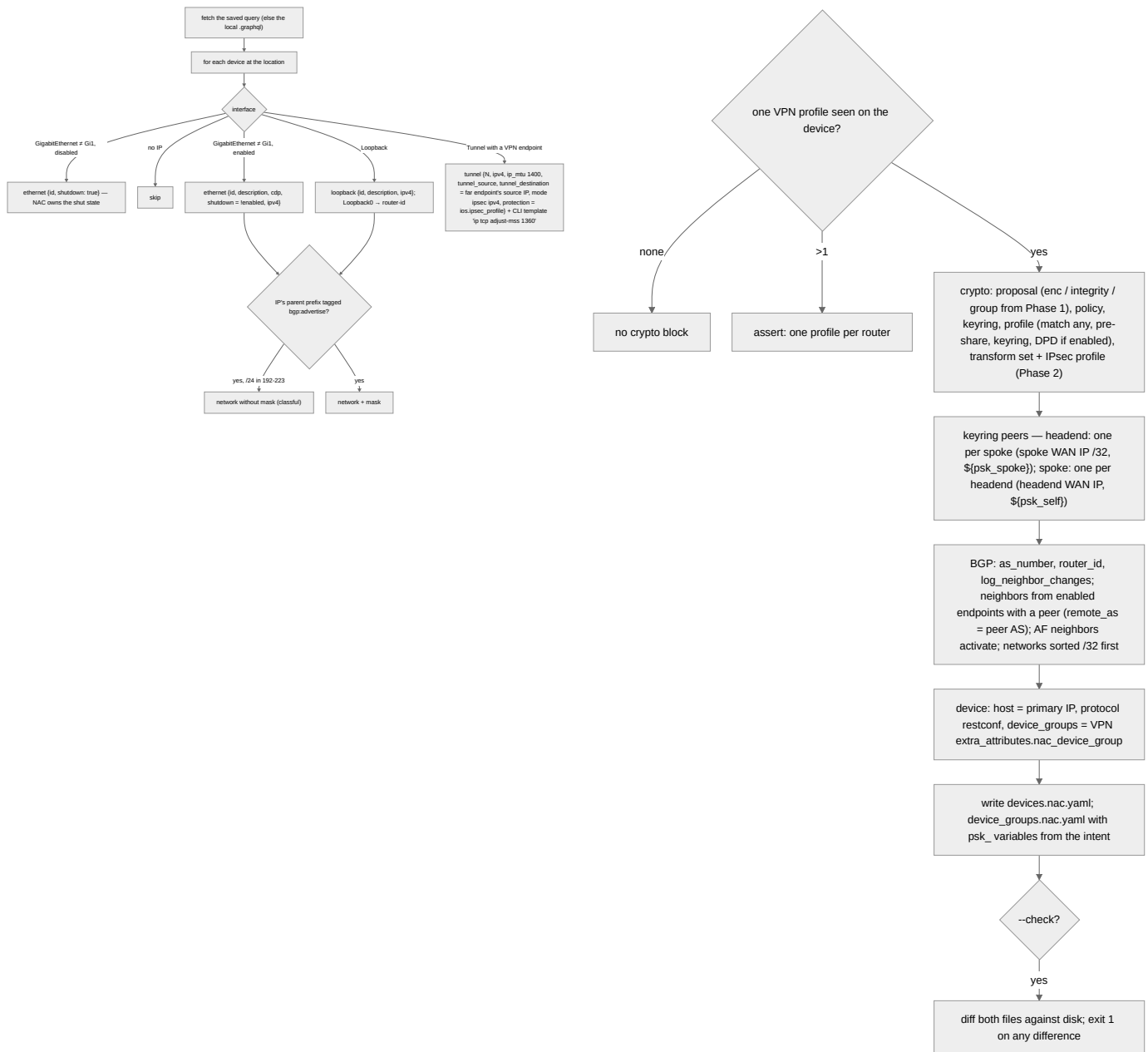
delete IPsec-VTI AS objects no device uses

saved GraphQL query: location + device names substituted; PATCH if the text differs

print N changes (0 on a re-run)

Comparison rule (`ensure()`): related objects compare by id, choice fields by value, MACs case-insensitively, JSON fields structurally (nested pynautobot Records serialised). Only fields that differ are PATCHed — a re-run reports **0 changes**.

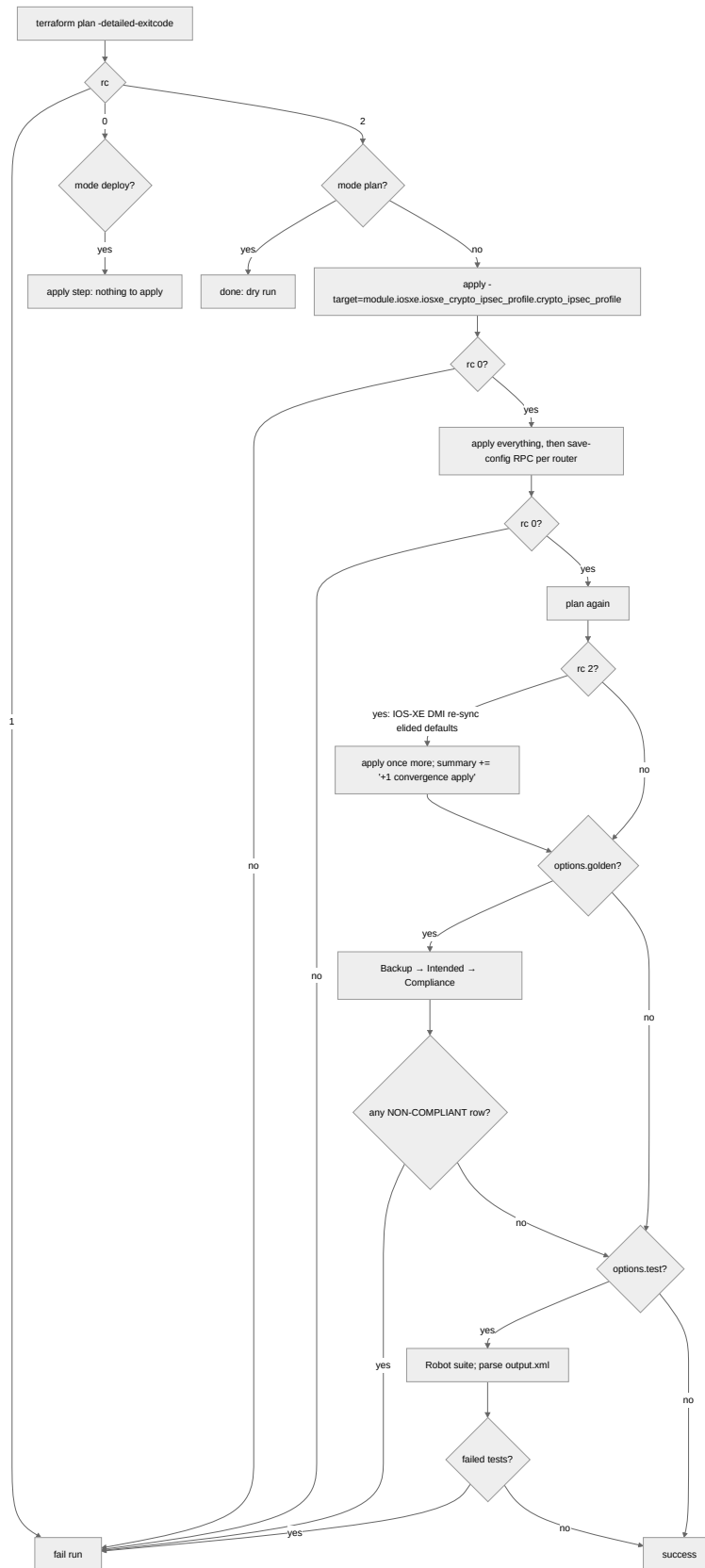
12. Decision trees: the renderer (`nautobot/render_nac.py`)



GraphQL renders single-choice fields as enum names (`IPSEC_TUNNEL` , `IKEV2`) but JSON-array choice fields raw (`AES-256-CBC`); `norm()` upper-cases and replaces "-" so both forms map to the same Cisco keywords.

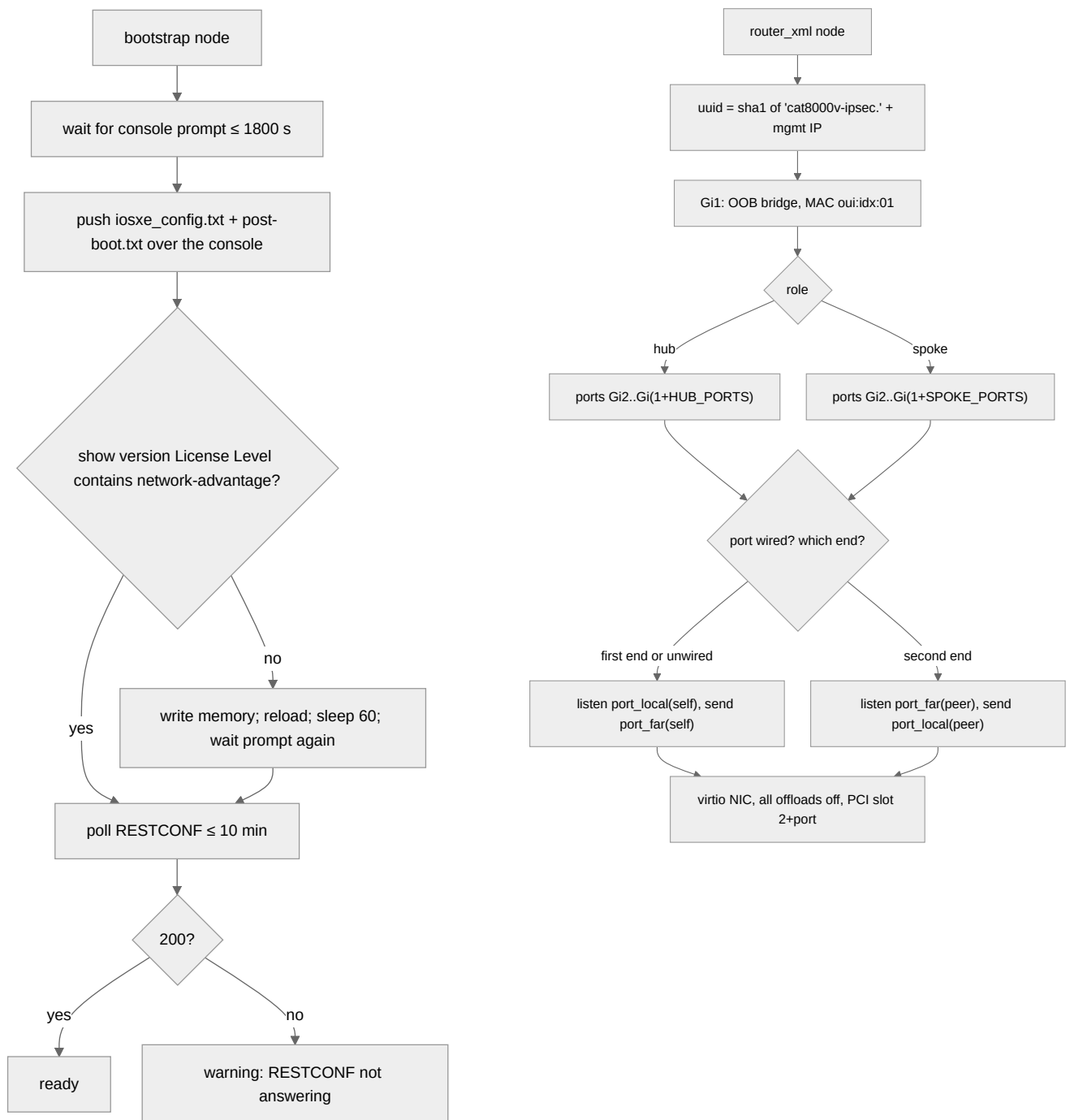
13. Decision trees: apply, convergence and Golden Config (`app.py`

`do_plan/do_apply/do_golden`)



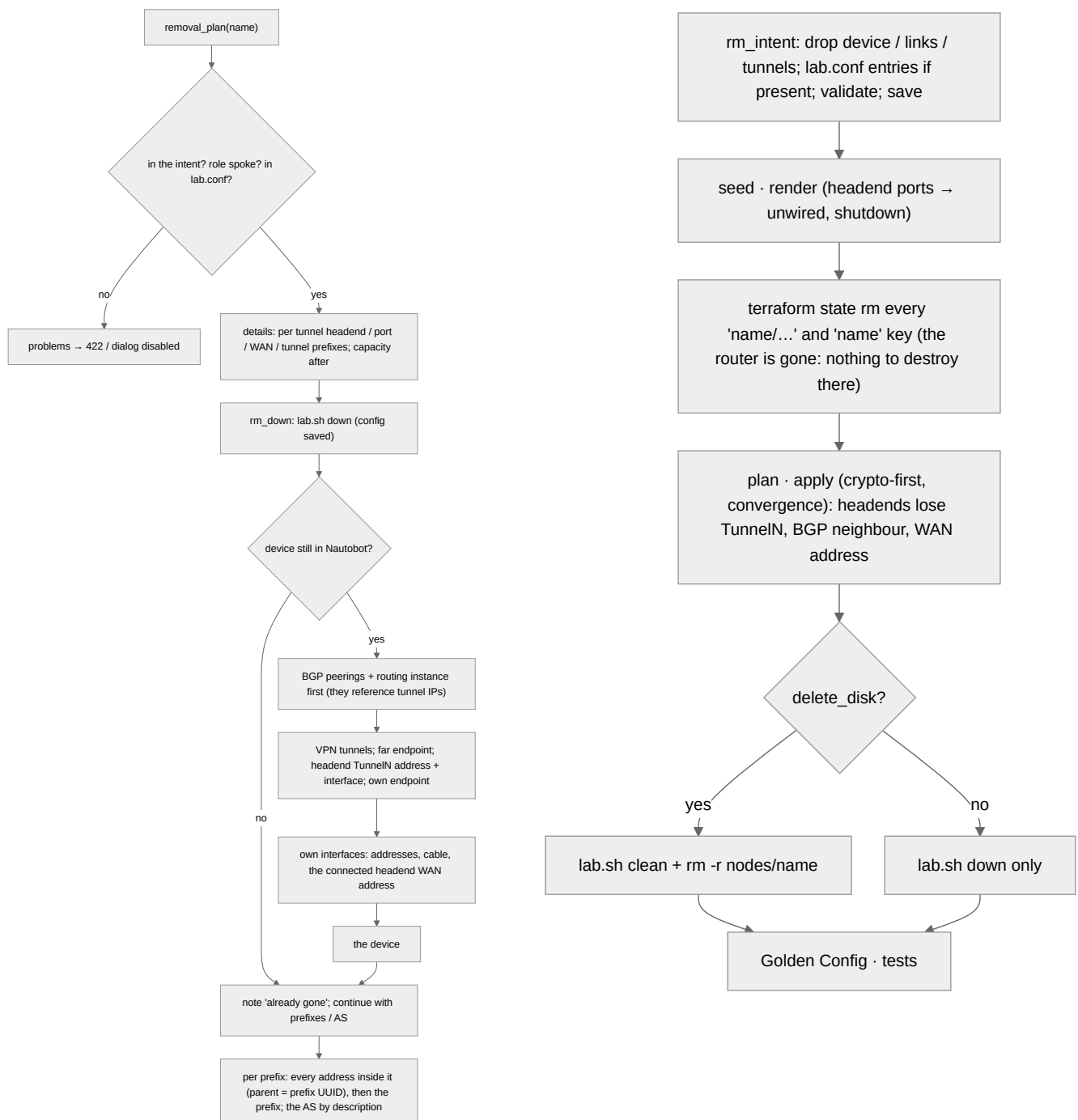
Why the crypto-profiles-first apply: the NAC module has no dependency from `iosxe_interface_tunnel` to `iosxe_crypto_ipsec_profile`, and IOS-XE refuses a tunnel whose `tunnel protection ipsec profile` does not exist yet — on a brand-new router the first full apply could push the VTI before its profile. **Why the convergence apply:** after an interface deletion or a reload IOS-XE re-syncs its YANG datastore and elides default values (transform-set key size, SSH defaults); one re-assert fixes it.

14. Decision trees: bootstrap and VM wiring (lab.sh)



Because a link's UDP pair is anchored on its first end (always the headend), adding a spoke never changes a headend's domain; adding a headend changes each spoke's definition (its new port's target), hence the spoke reboots in the hub workflow.

15. Decision trees: removal (`spokes.removal_plan` / `remove_from_nautobot`)



16. REST API reference (data in / data out)

Interactive documentation with schemas and examples: `/docs` (Swagger UI) and `/redoc`; OpenAPI JSON at `/openapi.json`.

Endpoint	Input	Output
GET <code>/api/intent</code>	—	intent, wiring (physical links), nodes (mgmt IP → VM name/idx/role/console), Nautobot public URL
GET <code>/api/inventory</code>	—	Nautobot devices at the lab location (name, mgmt IP, serial, model, role, status, URL), VPN URL
POST <code>/api/validate</code>	<code>{intent}</code>	<code>{problems: []}</code>
GET <code>/api/spokes/suggest?hubs=@ion=</code>	optional headends, region	SpokeSpec (all fields) + context (headends with region/distance/usage/free ports, regions, nearest, capacity)
GET <code>/api/hubs/suggest</code>	—	HubSpec + context (spokes to connect)
POST <code>/api/spokes/validate</code>	<code>{spoke: SpokeSpec}</code>	<code>{problems, hub_changes: {links: [per headend: interface, wan_ip, tunnel, tunnel_ip, tunnel_destination, bgp_neighbor, spoke{...}, tunnels_after, capacity], loopbacks}}</code>
GET <code>/api/spokes/{name}/removal</code>	name	<code>{problems, details: {mgmt_ip, asn, lan, router_id, links[...]}}</code>
POST <code>/api/runs</code>	<code>{mode, intent spoke hub, options{golden,test,delete_disk}}</code>	Run (id, mode, status, steps[name,title,status,started,finished,summary]); 409 if a run is active; 422 with problems
GET <code>/api/runs</code>	—	last 30 runs without logs (memory + <code>webapp/runs/*.json</code>)
GET <code>/api/runs/{id}?since=N</code>	log offset	Run + <code>log[{t,line}]</code> from N, <code>log_offset</code> , tests (suites → tests with status/message/elapsed), <code>results_dir</code> , devices, <code>resume_of</code>
POST <code>/api/runs/{id}/resume</code>	—	new Run with carried-over successes; 404/409
GET <code>/api/vpn-inventory?refresh=&live=</code>	flags	tunnels (model + live), headends (capacity, used, free, utilisation, firewall bandwidth, live CPU / QFP / DRAM, binding, aggregate, effective free, live status), devices (role, region, site, mgmt IP, AS, LAN, serial), vpns, summary, generated
GET <code>/api/vpn-inventory.csv</code>	—	CSV, one row per tunnel (35 columns)
<code>/results/...</code>	path	static Robot reports, logs, config backups

17. Test coverage map

Suite	Proves	Sources compared
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01 Management	SSH/RESTCONF reachability, hostname, IOS-XE version, Mgmt-vrf default route, license level, AAA/SSH/VTY hardening, banner	routers vs intent + NAC baseline
02 Underlay	every hub ↔ spoke p2p link up with the modelled /30s, ping both ways, no spoke ↔ spoke WAN reachability, CDP neighbour per link, loopbacks	routers vs intent
03 IPsec VTI	tunnel config lines, up/up, IKEv2 SAs READY with the modelled suite, DPD, ESP counters advance (encryption proven), sessions per headend = its tunnels	routers vs intent
04 Routing	eBGP Established headend ↔ spoke, spokes peer only with their headends, every site learnt, spoke-to-spoke AS path via a headend, LAN reachability, traceroute first hop is a headend	routers vs intent
05 NAC compliance	<code>terraform plan</code> exit code 0 (no drift)	routers vs NAC data
07 Firewalls	one VyOS firewall per headend reachable over OOB; interface addresses and LLDP neighbours as modelled; rendered configuration == running (<code>render_vyos.py --check</code>); only IKE/ESP/ICMP cross (ESP counters advance through the firewall), non-VPN traffic is dropped and logged; <code>firewall_bandwidth_mbps</code> equals the intent and the portal's capacity math (tunnels, bandwidth, CPU, aggregate) matches <code>intent.headend_capacity()</code>	firewalls vs Nautobot vs intent vs portal
06 Nautobot	devices/serials/locations, cables and /30s, VPN app objects and that each router's tunnel destination equals the far endpoint's source address, crypto policies vs routers, BGP model vs live sessions, <code>render --check</code> round-trip, Golden Config backups in Gitea and every compliance row compliant	Nautobot vs routers vs committed NAC data

18. File map

Path	Purpose
<code>lab.conf</code> , <code>lab.sh</code> , <code>tools/console.py</code> , <code>tools/vyos_console.py</code> , <code>tools/vyos_install.py</code> , <code>nodes/</code>	VM facts and libvirt controller (<code>up/down/bootstrap/rebuild/clean/rename/status/nac/nautobot/test/webapp</code>), IOS-XE and VyOS console automation, the one-off VyOS base-image installer, day-0 template and per-node files (<code>vyos_config.txt</code> for the firewalls)
<code>lab-intent.json</code> , <code>nautobot/intent.py</code>	The intent document; load/upgrade/validate/save, wiring & VM facts, region distance, nearest headends, PSK generation
<code>nautobot/onboard.py</code> , <code>seed.py</code> , <code>render_nac.py</code> , <code>render_vyos.py</code> , <code>golden_config.py</code> , <code>rename_node.py</code> , <code>nac-c8000v-ipsec-model.graphql</code> , <code>golden-config-templates/c8000v-ipsec.j2</code>	Nautobot integration: discovery, intent → objects, objects → NAC data, Golden Config setting/template/jobs, rename tool, the saved query, the Jinja template
<code>nac/main.tf</code> , <code>versions.tf</code> , <code>data/*.nac.yaml</code>	Terraform root: module + provider pins, baseline, generated device model and PSK variables
<code>tests/</code>	Robot suites 01–06, <code>LabLib.py</code> keywords, <code>lab_vars.py</code> (derived from the intent), <code>run.sh</code> , config capture
<code>webapp/app.py</code> , <code>schemas.py</code> , <code>spokes.py</code> , <code>inventory.py</code> , <code>static/index.html</code> , <code>runs/</code> , <code>demo/</code>	Portal API + pipeline runner, OpenAPI models, spoke/headend allocation and removal, inventory collector, UI, run records, demo recorders